

Note

NEVER EXPOSE THE PHOTOMULTIPLIER TUBE (PMT) TO ROOM LIGHTS WITH THE HIGH VOLTAGE TURNED ON. DO NOT EXCEED 1300 V.

Be sure to save your spectra in a way that you can access them! Save as ascii file.

Information

- Name:
- Lab Station:
- Lab Partner:
- Date:

1 Pre-Lab Conceptual Questions

1. What does the PMT do? Be sure to explain the purpose of both the photocathode and the dynodes.
2. Explain the process how we see a signal from a gamma-ray interaction in the scintillator. Try to include all the possible steps until we see a Gaussian shaped pulse from the shaping amplifier.
3. Why are photocathodes often curved ?
4. Draw the expected differential pulse height spectrum when exposing a NaI(Tl) scintillator to the 662-keV gammas from ^{137}Cs.
5. Two scintillation crystals are to be compared. The percent resolution, peak-to-total ratio, the full spectrum area and the photopeak area are of interest. Given the following properties of each of the scintillators, qualitatively compare them with respect to the four characteristics given above.

   Scintillator #1: NaI(Tl) 2" X 2" crystal
   Scintillator #2: BGO 3" X 3" crystal

2 Equipment Setup

2.1 Required Equipment

- PMT
- assorted scintilators
- amplifier
- HV power supply

3 Inorganic Scintillators

3.1 Tube Gain and Applied Voltage

In this section you will use a 2"x2" NaI(Tl) crystal. **Please do not expose bare PMT to room light under HV bias.**

- Assemble the crystal and PMT using optical coupling compound between the crystal and tube.
- Enclose the assembly in a light-tight container. Connect "POS HV" to the high voltage power supply and "anode" to the oscilloscope. Test for light leaks at a low voltage (about 600 V) by switching the room lights off and on while examining the system output on the scope. Look for changes in the baseline noise level of the PMT output signal as a sign of light leakage.
- Connect the PMT to an amplifier and the amplifer to MCA. Place a ^{137}Cs source near the crystal and observe the bright trace corresponding to the photopeak.
- Record the **photopeak channel number as a function of tube voltage** from the minimum voltage (~600 V) to 1300 V.
- Carefully watch the amplifier outputs on the scope as the voltage is raised, and be alert for the point at which the pulses begin to saturate (flat top).

NOTE: If the pulse exceeds saturation, drop the amplifier gain by a certain amount and multiply the measured channel number you are recording by that number.

3.2 Scintillator Pulse Height Spectra

- Set up the system to record the pulse height spectrum from the scintillator for a ^{137}Cs source. At a fixed operating voltage (around 800-900 V), adjust the amplifier gain so that the ^{137}Cs photopeak is at about ~45% of the full scale. Record the gain and high voltage you used. Keep the operating voltage and gain constant when using the different crystals later.
- Remove the source and take a 4-min background spectrum with no source. Background needs to be first subtracted before the following data analysis.
- Place the source as close as possible to the detector face and perform a 2-min measurement.
  - Set the MCA display to the logarithmic scale, and note the appearance of the sum peak at the upper end of the spectrum (you may not see the sum peak in some cases, but that is fine).
  - Save the spectrum in .SPE format.
  - Record/Calculate **the net area of the photopeak.** Use a reasonable method to determine the net area of the photopeak.
- With the source and detector in an easily reproducible geometry (~2 inches from the detector face), perform a 2-minute measurement.
  - Save the spectrum in .SPE format.
  - Measure/Calculate **the photopeak centroid, the photopeak energy resolution, the net photopeak counts and the total number of counts within the spectrum**.
- Repeat these (AT A FIXED GAIN) measurements for 1"x1" NaI 1) with and 2) without coupling compound, 3) 1"x1" BGO with coupling compound, 4) 1"x1" CsI(Tl) with coupling compound and 5) EJ-200 plastic scintillator with coupling compound.

Before changing detector crystals, turn off the high voltage (turn it to zero) !

4 Lab Report Requirement

While the report follows the usual structure, it must include include the answers to the following questions:

- Explain how you determined the net area counts of the peak. Do you think your method is reasonable?
- Describe all other recognizable features of the ^{137}Cs spectrum recorded from the 2"x2" NaI(Tl) crystal.
- Compare the differences in spectra (photopeak centroid, the photopeak energy resolution, the net photopeak counts and the total number of counts within the spectrum) among different scintillation crystals. Relate the different spectra to differences in the density, Z (atomic number), and physical size of the crystal.